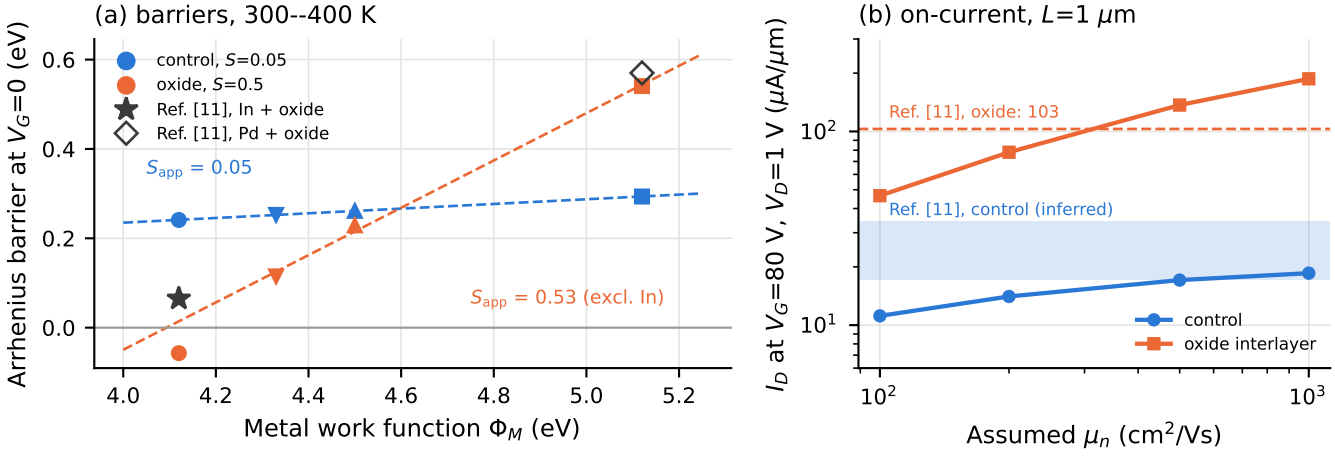
(a) barriers, 300--400 K
control, S=0.05
oxide, S=0.5
Ref. [11], In + oxide
Ref. [11], Pd + oxide
S_app = 0.05
S_app = 0.53 (excl. In)
Arrhenius barrier at V_G=0 (eV)
Metal work function Φ_M (eV)
0.0
0.2
0.4
0.6
4.0
4.2
4.4
4.6
4.8
5.0
5.2
(b) on-current, L=1 μm
Ref. [11], oxide: 103
Ref. [11], control (inferred)
control
oxide interlayer
I_D at V_G=80 V, V_D=1 V (μA/μm)
Assumed μ_n (cm²/Vs)
10¹
10²
10²
10³